

Math 22 HO1

2026-01-23

1.

(Lec 01 HOs. Exercises. 5)

$$\int_1^2 (x+1) \ln x \, dx$$

Solve for the indefinite integral. Use IBP with:

- $u = \ln x$, $dv = (x+1) \, dx$
- $du = \frac{1}{x} \, dx$, $v = \frac{x^2}{2} + x$

$$\begin{aligned} \int (x+1) \ln x \, dx &= \ln x \left(\frac{x^2}{2} + x \right) - \int \left(\frac{x^2}{2} + x \right) \frac{1}{x} \, dx \\ &= \ln x \left(\frac{x^2}{2} + x \right) - \int \left(\frac{1}{2}x + 1 \right) \, dx \\ &= \ln x \left(\frac{x^2}{2} + x \right) - \frac{1}{2} \int x \, dx + \int \, dx \\ &= \ln x \left(\frac{x^2}{2} + x \right) - \frac{1}{2} \left(\frac{x^2}{2} \right) + x \\ &= \ln x \left(\frac{x^2}{2} + x \right) - \frac{x^2}{4} + x \end{aligned}$$

Solve for the definite integral.

$$\begin{aligned} \int_1^2 (x+1) \ln x \, dx &= \left[\ln x \left(\frac{x^2}{2} \right) + x - \frac{x^2}{4} - x \right]_1^2 \\ &= \left[\ln 2 \left(\frac{4}{2} + 2 \right) - \frac{4}{4} - 2 \right] - \left[\ln 1 \left(\frac{1}{2} + 1 \right) - \frac{1}{4} - 1 \right] \\ &= (4 \ln 2 - 3) - \left(-\frac{1}{4} - \frac{4}{4} \right) \\ &= 4 \ln 2 - \frac{12}{4} + \frac{5}{4} \\ &= 4 \ln 2 - \frac{7}{4} \end{aligned}$$

2.

(Lec 02 HOs. Exercises. I. 7)

$$\int \frac{\sec^6(\ln x)}{x} dx$$

Substitute with $u = \ln x$, $du = \frac{1}{x} dx$:

$$\begin{aligned}\int \frac{\sec^6(\ln x)}{x} dx &= \int \sec^6 u \, du \\ &= \int \sec^2 u \sec^4 u \, du \\ &= \int \sec^2 u (\sec^2 u)^2 \, du \\ &= \int \sec^2 u (1 + \tan^2 u)^2 \, du\end{aligned}$$

Substitute with $v = \tan u$, $dv = \sec^2 u \, du$:

$$\begin{aligned}\int \frac{\sec^6(\ln x)}{x} dx &= \int (1 + v^2)^2 \, dv \\ &= \int (1 + 2v^2 + v^4) \, dv \\ &= \int dv + 2 \int v^2 \, dv + \int v^4 \, dv \\ &= v + \frac{2}{3}v^3 + \frac{v^5}{5} \\ &= \tan u + \frac{2}{3} \tan^3 u + \frac{1}{5} \tan^5 u \\ &= \tan(\ln x) + \frac{2}{3} \tan^3(\ln x) + \frac{1}{5} \tan^5(\ln x) + C\end{aligned}$$